

PROGRAM :

```
fact=1
n=int(input("Enter the number:"))
if n == 0:
    print("factorial of",n,"is:",fact)
else:
    for i in range(1,n+1):
        fact=fact*i
    print("factorial of",n,"is:",fact)
```

OUTPUT :

Enter the number:5
factorial of 5 is: 120

PROGRAM :

```
num = int(input("Enter the
number:"))
n1, n2 = 0, 1
print("Fibonacci Series:", n1, n2,
end=" ")
for i in range(2, num):
    n3 = n1 + n2
    n1 = n2
    n2 = n3
    print(n3, end=" ")
```

OUTPUT:

Enter the number:5
Fibonacci Series: 0 1 1 2 3

PROGRAM :

```
l1=[100,200,100]
print(l1)
sum=0
for i in range(0,len(l1)):
    sum=sum+l1[i]
print("Sum is :",sum)
```

OUTPUT:

[100, 200, 100]
Sum is : 400

PROGRAM :

```
from math import sqrt
n=int(input("Enter a limit:"))
for i in range(1000,n):
    if sqrt(i)== int(sqrt(i)) and i%2==0:
        print(i)
```

OUTPUT :

Enter a limit:1600
1024
1156
1296
1444

PROGRAM :

```
n=int(input("Enter the numbers:"))
for i in range(1,n+1):
    for j in range(0,i+1):
        a=i*j
        if a== 0:
            continue
        else:
            print(a,end=" ")
    print("\n")
```

OUTPUT:

Enter the numbers:5
1

2 4

3 6 9

4 8 12 16

5 10 15 20 25

PROGRAM :

```
def count(str):
    dict = {}
    for n in str:
        keys = dict.keys()
        if n in keys:
            dict[n] += 1
        else:
            dict[n] = 1
    return dict
print("Entr a string: ")
str = input()
print("Character frequency in string",str,"is",count(str))
```

OUTPUT :

```
Entr a string:
HAPPY BE HAPPY
Character frequency in string
HAPPY BE HAPPY is {'H': 2, 'A': 2, 'P':
4, 'Y': 2, ' ': 2, 'B': 1, 'E': 1}
```

PROGRAM :

```
st=input("enter the string")
if st.endswith ("ing"):
    st+='ly'
elif len(st)>=3:
    st+='ing'
print(st)

"""
string.endswith(string)
"""
```

OUTPUT :

```
enter the stringrun
runing
```

PROGRAM :

```
def longest_word(words):
    longest = 0
    for word in words:
        if len(word) > longest:
            longest = len(word)
    longest_word = word
    print("Longest word is",longest_word)
    print("Length of longest word is",longest)
    print("Enter a list of words separated by spaces: ")
    words = input().split()
    longest_word(words)
```

OUTPUT :

```
Enter a list of words
separated by spaces:
hello world programming
language
Longest word is
programming
Length of longest word is
11
```

PROGRAM :

```
n=int(input("Enter a limit"))
for i in range(0,n+1):
    for j in range(0,i):
        print("*",end=" ")
    print("\n")
for i in
reversed(range(0,n)):
    for j in
reversed(range(0,i)):
        print("*",end=" ")
    print("\n")
```

OUTPUT :

Enter a limit3

```
*  
  
* *  
  
* * *  
  
* *  
  
*
```

PROGRAM :

```
number=int(input(  
"Enter a number:  
"))
```

```
i=1
```

```
print("The factors  
of",number,"are:")  
while i<=number:  
    if number%i ==  
0:    print(i,end="  
")  
    i=i+1
```

OUTPUT :

Enter a number:12

The factors of 12
are:

1 2 3 4 6 12

PROGRAM :

```
square=lambda x:x*x  
rect=lambda l,b:l*b  
tri=lambda b,h:0.5*b*h  
a=int(input("Enter the side of square : "))  
b=int(input("Enter length of rectangle : "))  
c=int(input("Enter breadth of rectangle : "))  
d=int(input("Enter base of traingle : "))  
e=int(input("Enter height of traingle : "))  
print("Area of square : ",square(a))  
print("Area of rectngle : ",rect(b,c))  
print("Area of triangle : ",tri(d,e))
```

OUTPUT :

Enter the side of square : 5

Enter length of rectangle :6

Enter breadth of rectangle
:4

Enter base of traingle : 5

Enter height of traingle : 6

Area of square : 25

Area of rectngle : 24

Area of triangle : 15.0

